PRESYOSERBISYO · DTI CATANDUANES

ARIMA Price Forecasting Method

How the system projects next week's commodity prices: the formula, the steps, and a short explanation. November 2026.

1 In short

For each commodity, PresyoSerbisyo looks at the prices monitoring officers have recorded in stores and projects the next 7 values. It uses **ARIMA**, a standard statistical method for forecasting a value from its own history. The "Forecasted next week" figure on the Price Trends page is the 7th projected value. The forecast is an **estimate based on past prices**, not a guarantee. It cannot anticipate events that never appeared in the price history, such as a new SRP bulletin, a typhoon or a supply shortage.

2 What ARIMA means

ARIMA stands for **AutoRegressive Integrated Moving Average**. A model is written ARIMA(p, d, q), where each number switches on one part:

Part	Order	What it does
AutoRegressive (AR)	p	Predicts the next price from the last p prices.
Integrated (I)	d	Works on price *changes* instead of prices, differenced d times, to remove a steady upward or downward drift.
Moving Average (MA)	q	Corrects the prediction using the last q prediction errors.

GENERAL ARIMA MODEL

$$y'_t = c + \varphi_1 y'_{t-1} + \ldots + \varphi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

where y'_t is the price series after differencing d times, c is a constant, φ are the autoregressive coefficients, θ are the moving-average coefficients, and ε_t is the random error, the part of the price that the history cannot explain.

3 The model PresyoSerbisyo uses: ARIMA(1, 0, 0)

The system uses the simplest form: **one** autoregressive term, **no** differencing and **no** moving-average term ($p = 1$, $d = 0$, $q = 0$). With those orders, the general formula reduces to:

THE MODEL USED

$$y_t = c + \varphi \cdot y_{t-1} + \varepsilon_t$$

Each price is a constant plus a fraction (φ) of the previous price, plus some unpredictable noise. The values of c and φ are estimated automatically from the commodity's own recorded prices every time a forecast is generated.

Projecting forward

Future noise is unknown, so it is set to zero, and each projected value feeds into the next one:

FORECAST, STEP BY STEP

$$\hat{y}_{t+1} = c + \varphi \cdot y_t \qquad \hat{y}_{t+h} = c + \varphi \cdot \hat{y}_{t+h-1} \qquad (h = 2, 3, \ldots, 7)$$

The same thing can be written directly in terms of the long-run average price μ:

EQUIVALENT FORM

$$\hat{y}_{t+h} = \mu + \varphi^h \cdot (y_t - \mu), \quad \text{where } \mu = c \,/\, (1 - \varphi)$$

Plain reading: the forecast starts at the latest recorded price and moves step by step toward the commodity's usual (average) price. φ controls how fast it moves. A value close to 1 means prices are "sticky" and the forecast stays near today's price. A value closer to 0 means it returns to the average quickly.

> **Worked example.** Suppose a commodity's usual price is μ = ₱45.00, the latest recorded price is y_t = ₱46.00, and the estimated φ = 0.6. Then:
> Step 1: $45.00 + 0.6 \times (46.00 - 45.00) = ₱45.60$
> Step 2: $45.00 + 0.6^2 \times (46.00 - 45.00) = ₱45.36$
> ...
> Step 7: $45.00 + 0.6^7 \times (46.00 - 45.00) \approx ₱45.03$
> The forecast drifts from ₱46.00 back toward the usual ₱45.00. The figures are illustrative, not taken from a real commodity.

4 How it works in the system

1. **Collect.** All recorded prices for the commodity, from every monitored store, are arranged from oldest to newest.
2. **Clean.** Missing or non-positive values are removed.
3. **Check the amount of data.** At least **3** recorded prices are needed. With fewer, no forecast is shown.
4. **Choose the method.**
   - **8 or more prices:** the ARIMA(1, 0, 0) model above is fitted and projects 7 steps ahead.
   - **3 to 7 prices:** too few to fit ARIMA reliably, so the system extends the recent average change instead (the trend formula below).
   - If ARIMA cannot be fitted, or returns an invalid result such as a zero or negative price, the system falls back to the trend formula.
5. **Store and show.** The 7 values are rounded to the centavo and dated one day apart, starting the day after the latest recorded price. The 7th value is displayed as "Forecasted next week".
6. **Refresh.** The forecast is recalculated each time a commodity's forecast is viewed, so it always reflects the latest recorded prices.

FALLBACK TREND FORMULA (3–7 PRICES)

$$\hat{y}_{t+h} = y_t + h \cdot \bar{\Delta}$$

where $\bar{\Delta}$ is the average change between consecutive recorded prices. The latest price is simply extended by that average change for each step.

5 Confidence level

Each projected value carries a confidence score from 0.20 to 0.99, shown as **High**, **Medium** or **Low**. It is a **rule-of-thumb indicator of how stable and well-supported the price history is**, not a statistical probability:

CONFIDENCE SCORE FOR FORECAST STEP K

$$\text{score} = 0.90 - 1.2 \cdot v - h_k + s \quad \text{(kept between 0.20 and 0.99)}$$

Term	Meaning	Formula
Volatility v	How much prices swing relative to their average. More swing gives lower confidence.	v = standard deviation ÷ average price
Data support s	More recorded prices give higher confidence, up to a cap.	$s = \min(0.25, n \div 100)$
Horizon penalty h_k	Further-ahead steps are less certain.	$h_k = \min(0.15, 0.03 \times k)$

High = 0.75 or above · **Medium** = 0.50 to 0.74 · **Low** = below 0.50. The Price Trends page shows the confidence of the 7th step.

6 Limitations

- The forecast learns only from past recorded prices. Outside events, such as SRP revisions, weather or supply problems, are not part of the model.
- Prices from all monitored stores are combined into one sequence. Each forecast step is one recorded price ahead and is labelled as one day.
- ARIMA(1, 0, 0) favours a return toward the usual price. It does not extend a long upward or downward trend, and it does not model seasonal patterns.
- Forecasts improve as more prices are recorded. Commodities with few records use the simpler trend formula.
